Gunakarthik Naidu Lanka

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Professional Summary

MS Computer Science candidate (4.0 GPA, Dec 2025) specializing in Generative AI, Conversational AI agents, and cloud-based backend systems. Experienced building production AI applications with Python, SQL, Vertex AI, Google Cloud APIs, BigQuery, and Looker dashboards. Strong foundation in microservices architecture, ETL pipelines, and data migration. Built HireAgent: Conversational AI agent processing 800+ applications at 98% task completion using Vertex AI, multi-LLM routing, and intelligent prompt engineering.

Technical Skills

Languages: Python, SQL (PostgreSQL, BigQuery), Java, Go, TypeScript, JavaScript, C++

AI/ML: Generative AI, Conversational AI, Vertex AI, Dialogflow, LLM Integration, Prompt Engineering, RAG, NLP, AI Agents

Google Cloud: Vertex AI, BigQuery, Cloud APIs, Looker, GCP Storage, Cloud Functions

Cloud & Infra: AWS (EC2/S3/Lambda), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Linux

Backend: FastAPI, Spring Boot, Node.js, REST APIs, gRPC, Microservices, Event-Driven Architecture, Pub/Sub

Data & ETL: BigQuery, Looker, Tableau, ETL Pipelines, Data Migration, SQL Optimization, Schema Design, Data Quality

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Vector Databases

Concepts: Distributed Systems, System Design, Agile/Scrum, Performance Optimization, Unit Testing, Jira, Git

Professional Experience

Software Engineering Intern, Velocity Tech *(Zinio TalentHub, AI-Driven Hiring Platform)* — Jan 2024 – Dec 2024

- Engineered AI-powered backend microservices using Python, FastAPI, and Spring Boot integrating Generative AI for automated candidate analysis; maintained 99.9% uptime under high-volume concurrent load.
- Designed normalized PostgreSQL schemas with advanced indexing achieving 45% reduction in p99 retrieval latency; built ETL pipelines for data migration and quality validation across 500+ candidate records.
- Built REST APIs across 10+ endpoints; deployed containerized microservices via Docker and CI/CD to AWS with automated testing and rollback support; led prompt engineering cycles improving AI accuracy by 30%.

Full Stack Engineer, EPICS at ASU *(Foster Arizona Community Platform)* — Aug 2023 – Dec 2023

- Architected dashboard interfaces with React.js and Python REST APIs integrating real-time data workflows for intelligent resource discovery serving Arizona foster youth, reducing search time by 40%.
- Containerized services with Docker and deployed to AWS (EC2/S3); engineered comprehensive testing framework with 95% code coverage eliminating critical data integrity bugs pre-launch.

Teaching Assistant, Arizona State University *(Data Structures & Algorithms)* — Jan 2023 – Dec 2024

- Mentored 100+ students across Python and C++ in data structures, algorithms, and software engineering fundamentals; maintained 92% student satisfaction across 4 semesters.
- Designed assignments bridging academic theory with production practice, reducing student implementation error rates by 35% through structured debugging workshops and time-complexity deep-dives.

AI & Cloud Engineering Projects

HireAgent *(Conversational AI Agent for Job Applications)* — *Python, Vertex AI, BigQuery, Looker, GCP*

- Built end-to-end Conversational AI agent processing 4,000+ applications using Vertex AI with Generative AI models (Claude, Gemini, DeepSeek) for intelligent multi-LLM routing; achieved 98% task completion across 5 ATS platforms.
- Engineered Google Cloud APIs and Vertex AI pipeline for automated decision-making with prompt engineering optimization; designed BigQuery analytics layer and Looker dashboards for real-time ATS scoring and monitoring.

NeighborhoodPulse *(Multi-Agent Geo-Spatial Intelligence)* — *Python, Vertex AI, BigQuery, Looker, LangGraph, GCP*

- Engineered multi-agent Conversational AI using Vertex AI and LangGraph; integrated Generative AI APIs for structured insight generation with 94% citation accuracy.
- Built BigQuery ETL pipeline aggregating geospatial data (FEMA, zoning, climate); designed Looker dashboards reducing manual research time by 85%.

Vin-Sight *(Vehicle Intelligence Conversational Agent)* — *Python, Vertex AI, Dialogflow, Llama 3, Phi-3, NHTSA API*

- Architected privacy-preserving Conversational AI engine using Vertex AI and Dialogflow concepts for natural language VIN decoding and federal risk auditing; applied GGUF quantization achieving 2.3× throughput improvement on 8GB VRAM.

Education

Arizona State University, Tempe, AZ — **GPA: 4.0/4.0**

MS in Computer Science — Dec 2025

BS in Computer Science, Minor in Business — Dec 2024